

NASIK, India

WMO: 429210

Lat: 19.967N

Lon: 73.817E

Elev: 1962

StdP: 13.68

Time Zone: 5.50 (IND)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 48.5	(c) 51.1	(d) 30.1	(e) 26.0	(f) 82.8	(g) 33.5	(h) 30.1	(i) 81.4	(j) 7.6	(k) 79.1	(l) 7.0	(m) 77.7	(n) 0.1	(o) 320	(p) 0.254

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	23.7	100.8	67.0	99.1	67.1	97.2	67.3	77.0	85.5	76.3	84.1	75.6	82.9	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	141.7	79.7	74.4	138.3	78.9	73.8	135.3	78.1	42.1	85.6	41.4	84.4	40.7	83.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.9	(b) 7.8	(c) 7.1	(e) 43.8	(f) 104.4	(g) 2.7	(h) 1.6	(i) 41.8	(j) 105.5	(k) 40.2	(l) 106.5	(m) 38.7	(n) 107.4	(o) 36.8	(p) 108.6		
			DB													
			WB													
				40.5	79.8	3.6	1.7	37.9	81.0	35.8	81.9	33.8	82.9	31.2	84.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.5	67.6	72.1	78.5	84.2	85.4	81.7	76.7	75.8	76.9	76.8	72.9	69.2
	DBStd	6.13	3.52	4.14	4.30	3.13	2.03	3.14	2.10	1.69	2.03	2.64	3.33	3.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	31	16	6	0	0	0	0	0	0	0	0	0	9
	CDD50	9674	547	620	885	1026	1099	951	829	799	806	831	686	595
	CDD65	4231	98	206	420	576	634	501	364	334	356	366	236	140
	CDH74	41175	1342	2404	5231	7691	8053	4767	1731	1243	1939	3113	2199	1462
	CDH80	18607	325	1060	2983	4720	4629	1821	235	109	428	1229	719	349
(14) Wind	WSAvg	2.2	1.2	1.6	1.8	2.5	3.9	3.7	3.3	3.0	1.9	1.3	1.3	1.3
(15) Precipitation	PrecAvg	26.9	0.1	0.0	0.1	0.2	0.7	4.0	7.3	5.5	6.5	1.7	0.8	0.3
	PrecMax	45.1	0.9	0.4	0.7	1.7	3.9	11.5	17.2	13.3	16.8	5.5	4.3	3.8
	PrecMin	15.3	0.0	0.0	0.0	0.0	0.0	0.0	2.5	1.7	0.7	0.0	0.0	0.0
	PrecStd	6.7	0.2	0.1	0.2	0.4	1.1	3.3	3.8	3.0	4.2	1.7	1.2	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	94.4	101.0	103.5	102.5	97.1	87.6	84.8	89.1	91.6	89.1	88.0
		MCWB	62.6	63.0	64.5	65.8	69.9	74.5	74.7	75.4	74.6	69.5	66.7	65.1
	2%	DB	85.4	91.7	98.3	101.4	100.5	94.4	84.7	82.8	86.6	89.7	87.5	85.6
		MCWB	62.0	62.5	64.4	66.0	69.0	74.0	74.6	74.6	74.7	69.7	66.7	64.8
	5%	DB	83.6	89.2	95.9	99.7	98.7	91.7	82.6	81.2	84.3	88.2	86.1	83.7
		MCWB	61.6	62.1	63.6	65.9	69.2	74.3	74.3	74.0	74.5	69.0	66.5	64.3
	10%	DB	81.6	86.9	93.7	97.8	96.7	89.0	80.8	79.7	82.5	86.7	84.5	81.8
		MCWB	61.1	61.7	63.5	65.8	69.9	74.6	74.2	73.6	74.0	68.8	66.3	63.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.2	68.0	72.6	73.8	76.9	78.1	76.8	76.6	78.2	76.9	74.2	71.1
		MCDB	80.4	85.6	89.5	89.7	92.4	87.7	82.5	82.0	84.2	84.1	82.8	81.4
	2%	WB	65.0	65.9	68.8	71.4	75.3	77.1	75.6	75.6	76.5	75.5	73.0	69.0
		MCDB	79.6	83.8	87.1	86.7	89.4	86.6	80.7	80.4	82.8	82.7	81.1	79.9
	5%	WB	63.8	64.5	66.8	70.1	74.4	76.3	75.1	74.7	75.6	74.5	71.8	67.4
		MCDB	79.0	81.5	86.6	86.8	87.7	85.4	79.9	79.1	81.3	81.3	78.8	78.6
	10%	WB	62.5	63.1	65.4	68.9	73.6	75.6	74.5	74.1	74.7	73.5	70.5	65.9
		MCDB	77.7	81.3	85.8	86.9	86.1	84.2	78.9	78.1	79.9	80.0	77.3	77.1
(35) Mean Daily Temperature Range	MDBR		28.9	29.5	29.7	28.1	23.7	13.7	7.5	7.6	11.5	20.3	24.8	26.9
	5% DB	MCDBR	30.6	32.1	31.4	29.5	26.6	18.7	11.0	10.3	15.0	23.5	26.5	28.6
		MCWBR	11.1	9.5	8.4	7.7	7.4	4.0	3.8	4.2	5.8	7.2	9.2	10.8
	5% WB	MCDBR	26.1	27.4	27.5	27.0	22.2	14.7	8.3	8.8	12.4	16.0	17.0	23.0
		MCWBR	10.6	9.4	9.5	8.7	6.2	4.8	3.7	4.2	5.7	6.5	7.2	9.9
(40) Clear-Sky Solar Irradiance	taub		0.418	0.423	0.470	0.510	0.526	0.595	0.585	0.548	0.530	0.531	0.485	0.444
	taud		2.159	2.152	2.032	1.971	1.972	1.862	1.895	1.998	2.029	1.954	2.027	2.088
	Ebn at Noon		269	276	267	257	250	231	234	244	247	241	246	255
	Edh at Noon		44	47	56	60	59	65	63	57	55	57	50	47
(44) All-Sky Solar Radiation	RadAvg		1534	1801	2086	2261	2268	1574	1093	1104	1404	1652	1533	1460
	RadStd		44	53	68	47	98	160	136	90	126	90	82	51

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.65	+1.82	N/A	N/A	N/A	N/A	+0.38	N/A	N/A	+288	+290

Nomenclature: See separate page